Vardhan Palod

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Research Objective: My current research focuses on understanding and enhancing the reasoning abilities of Large Language Models (LLMs). In particular, my work explores: (i) the semantic properties and underlying causes of performance improvements arising from the use of intermediate tokens in language models, and (ii) the effects of reinforcement learning on task performance and the interpretability of intermediate reasoning processes. Additionally, I am actively developing robust and scalable frameworks to enable effective and trustworthy Human–AI collaboration.

Education

Ira A. Fulton School of Engineering, Arizona State University — *Aug 2025 – May 2030 (Expected)*
PhD in Computer Science, Advisor: *Prof. Subbarao Kambhampati* — GPA: 4.0/4.0

Ira A. Fulton School of Engineering, Arizona State University — *Aug 2024 – May 2025*
Master of Science in Computer Science - converted to PhD — GPA: 4.0/4.0

Birla Institute of Technology and Science, Goa, India — *Aug 2020 – May 2024*
B.E. in Computer Science with a minor in Data Science — CGPA: 8.56/10

Research Experience

Graduate Research Associate: School of CS & AI, ASU — *Jan 2025 - Present*
Mentored by *Dr. Subbarao Kambhampati*

- Investigating language and reasoning models' reasoning abilities via fine-tuning, and RL-training and developing constructive applications of LLMs as complementary tools alongside the robust planners developed in the AI Planning community.

Master's Opportunity for Research in Engineering (MORE), ASU — *Jan 2025 - May 2025*
Mentored by *Dr. Heewook Lee*

- Fine-tuned RITA (protein language model) for generating binding high-quality T-cell receptor sequences (TCRs) for epitopes (part of the antigen that triggers immune response) which had no known binding TCRs. The fine-tuned model achieved 91.91% good TCR generation rate with strong predicted binding affinity scores, demonstrating its potential for novel immunotherapy design.

Publications

- **Beyond semantics: The unreasonable effectiveness of reasonless intermediate tokens.**
  Kaya Stechly*, Karthik Valmeekam*, **Vardhan Palod***, Atharva Gundawar*, Subbarao Kambhampati. (*Equal contribution.)
  NeurIPS 2025, Foundations of Reasoning in Language Models Workshop.
  NeurIPS 2025, Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning (LAW).
- **Performative Thinking? The Brittle Correlation Between CoT Length and Problem Complexity.**
  Vardhan Palod, Karthik Valmeekam, Kaya Stechly, Subbarao Kambhampati.
  NeurIPS 2025, Workshop on Efficient Reasoning.
- **Who is Helping Whom? Analyzing Inter-dependencies to Evaluate Cooperation in Human-AI Teaming.**
  Upasana Biswas, **Vardhan Palod**, Siddhant Bhambri, Subbarao Kambhampati.
  Second Coordination and Cooperation in Multi-Agent Reinforcement Learning Workshop, 2025 (Oral Presentation).
- **Stop Anthropomorphizing Intermediate Tokens as Reasoning/Thinking Traces!**
  Subbarao Kambhampati, Kaya Stechly, Karthik Valmeekam, Lucas Saldyt, Siddhant Bhambri, **Vardhan Palod**, Atharva Gundawar, Soumya Rani Samineni, Durgesh Kalwar, Upasana Biswas.
  NeurIPS 2025, Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning (LAW).
  NeurIPS 2025, Workshop on Interpreting Cognition in Deep Learning Models (CogInterp).
- **Discounting and Drug Seeking in Biological Hierarchical Reinforcement Learning.**
  Vardhan Palod, Pranav Mahajan, Veeky Baths, Boris S. Gutkin.
  CCN 2025 (Computational Cognitive Neuroscience Conference).

Notable Research Projects

Inference time scaling of LLMs with Rollouts Jan 2025 - Present

- Built an inference-time scaling technique for large language models (LLMs) enabling strategic lookahead through rollouts before committing to the next action in reasoning tasks. Evaluated GPT-4o-mini on the Game of 24 domain, achieving 70% accuracy, a 3x improvement over the prior 21% baseline.

Leveraging LLMs as World-models for training RL Agents Aug 2024 - Dec 2024

- Developed a framework using Llama 3.1 to generate state-action trajectories for Gridworld environments, enabling accelerated convergence of model-free RL agents to optimal policies by up to 50% and minimized unsafe exploration by up to 75% as compared to agents that are directly trained in real environments

Technical Strengths

Programming Languages: Python, C/C++, JAVA, PDDL

Frameworks: verl (GRPO and FSDP SFT), Pytorch, HuggingFace, Docker, Kubernetes, Scikit-Learn, Numpy, Pandas

Teaching & Service

Teaching:

- **Teaching Assistant - CSE 355** Jan 2025 - May 2025

  Introduction to Theoretical Computer Science by Prof. Heekwook Lee

Reviewing:

- **(AAAI 26)** *The fortieth Annual AAAI Conference on Artificial Intelligence*
- **(NeurIPS 25 LAW workshop)** *Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning*